Computer Science & IT

COMPILER DESIGN
(CD)

Parsers

Lecture No. 5



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

First

Topics to be Covered



- Topic
- Topic
- Topic

First and follow



Extra Ordinary Individuals: Stories to Ignite Student Motivation

Mahendra Singh Dhoni (b. 1981) captained India to victory in the 2007 T20 World Cup, 2011 Cricket World Cup, and 2013 Champions Trophy. Known for his cool demeanor under pressure and innovative finishes, 'Captain Cool' rose from railway-station grounds man to one of cricket's greatest captains. He mentors young players and invests in sports academies.

Lesson: Composed leadership and tactical innovation can guide teams to triumph in high-stakes arenas.



Extra Ordinary Individuals: Stories to Ignite Student Motivation

Sachin Tendulkar (b. 1973) debuted for India at age 16, becoming the highest run-scorer in international cricket history. His 100 international centuries and unwavering dedication earned him national reverence. Tendulkar's disciplined work ethic, sportsmanship, and pursuit of excellence inspired generations of athletes worldwide.

Lesson: Lifelong dedication and humility in success build enduring legacies that motivate others.



Extra Ordinary Individuals: Stories to Ignite Student Motivation

Milkha 'The Flying Sikh' Singh (1929–2021) overcame the trauma of Partition as an orphan to become India's first Commonwealth Games gold medalist in 1958 and narrowly missed an Olympic medal in 1960. His 400m records and unyielding training regimen made him a national icon. Singh later trained young athletes and advocated sports development.

Lesson: Channeling personal tragedy into disciplined pursuit can propel one to historic athletic achievements.



Extra Ordinary Individuals: Stories to Ignite Student Motivation

P. T. Usha (b. 1964) was India's greatest female sprinter, winning multiple Asian Games golds and missing the 1984 Olympic medal by 0.01 seconds in the 400m hurdles. Known as the "Payyoli Express" she dominated track events in the 1980s and later founded a sports academy to train budding athletes.

Lesson: Competitive spirit and unwavering training amid narrow margins can elevate performance to near perfection.

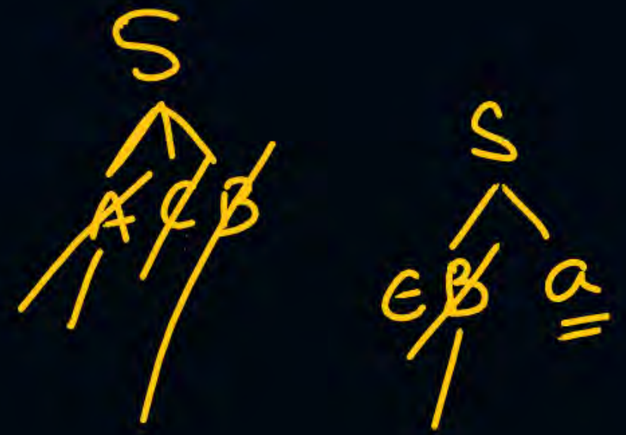
	first
$E \to TE'$	$\{id, ($
$E' \to +TE' / \epsilon$	$\{+, \epsilon\}$
$T \to FT'$	$\{id, ($
$T' \to *FT' / \epsilon$	$\{*, \epsilon\}$
$F \to id / (E)$	$\{id, ($

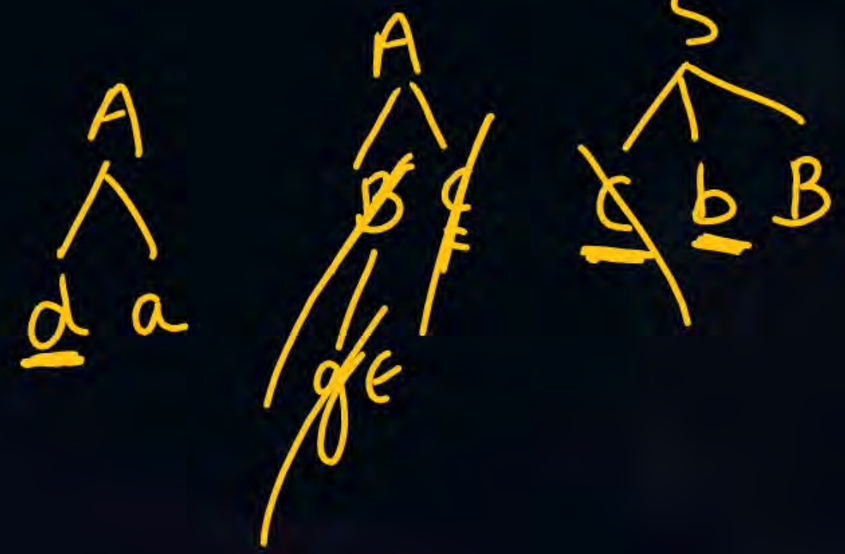
	first
$S \to ACB / CbB / Ba$	$\{d, g, h, \epsilon, b, a\}$
$A \to da / BC$	$\{d, g, h, \epsilon\}$
$B \to g / \epsilon$	$\{g, \epsilon\}$
$C \to h / \epsilon$	$\{h, \epsilon\}$



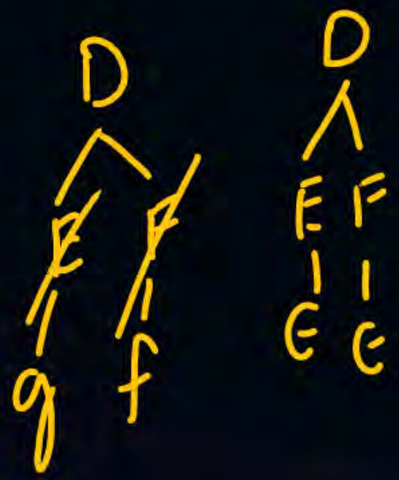
	first
$S \rightarrow aABb$	$\{a\}$
$A \rightarrow c/\epsilon$	$\{\epsilon, c\}$
$B \rightarrow d/\epsilon$	$\{\epsilon, d\}$

	First
$S \to ACB / CbB / Ba$	$\{d, g, h, \epsilon, b, a\}$
$A \to da / Bc$	$\{d, g, h, \epsilon\}$
$B \to g / \epsilon$	$\{g, \epsilon\}$
$C \to h / \epsilon$	$\{h, \epsilon\}$





$S \to aBDh$	$\{a\}$
$B \to cC$	$\{c\}$
$C \to bC/\epsilon$	$\{b, \epsilon\}$
$D \to EF$	$\{g, f, \epsilon\}$
$E \to g/\epsilon$	$\{g, \epsilon\}$
$F \to f/\epsilon$	$\{f, \epsilon\}$



Follow(A) where A is non terminal :-

Follow(A) gives set of all terminals that may appear immediately to the right of 'A' in some sequential form. (intermediate string)

If 'S' is the start symbol and $ is i/p right end marker then place $ in Follow(A).

$E \to E+E / E*E / id$

$E\$$ ✓

$E+E\$$

$E*E+E\$$

$E\$$ ✓

Follow(E) = { +, *, $ }

Follow(A) does not contain ε.

$S \rightarrow AE$

$E \rightarrow \in$

$S\,\$$ ✓

$AE\,\$$ ✓

$A\in\,\$$ ✓

$A\,\$$ ✓

When ever we start derivation in TDP, we always start with $S\$$. where $ is a special symbol which indicates end of the string

Rules to find follow(A) :-

→ If there is a production $B \to \alpha A \beta$, where β is a string of grammar symbols, then First(β) except 'ϵ' is placed in follow(A)

→ If there is a production $B \to \alpha A$ or a production $B \to \alpha A \beta$ where First(β) contains 'ϵ', then everything in follow of B is in follow(A).

$B \to \alpha A \beta$

$B \to \alpha A$

$E \to E+E / E*E$ ✓ ✓

E $

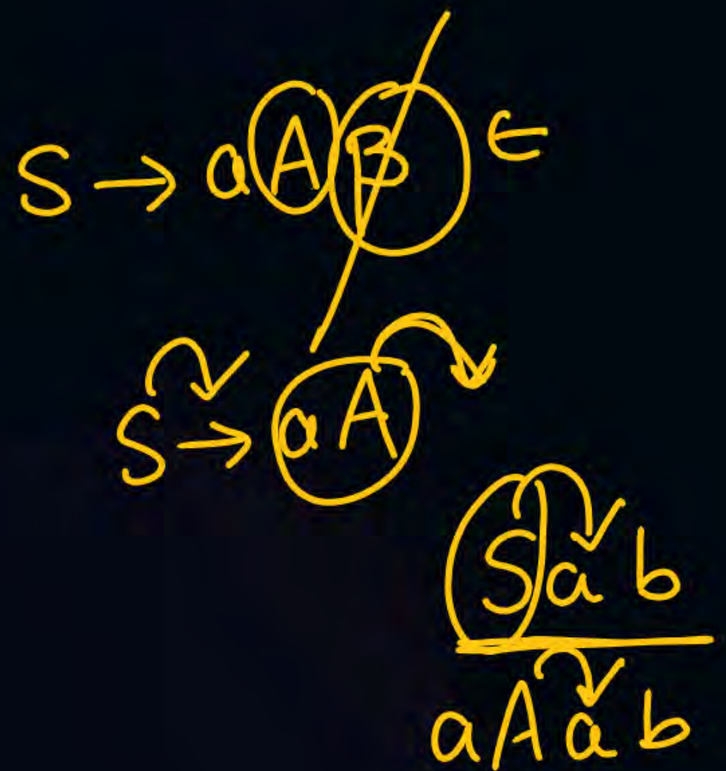
E+E $

E*E+E $

{ $, +, * }

B a c

a A a c


S → aAB ∈
S → aA
Sab
aAab

Note:- In Add to find the follow, see the RHS of productions

$A \rightarrow aBB$

$A \rightarrow aB$ / ~~$B \in$~~

$A \rightarrow aB$

	follow	first
S → ABCD	{$}	{b, d}
A → b/ϵ	{d}	(b, ϵ)
B → C	{d}	d
C → d	{e}	d
D → ϵ	{$}	ϵ

follow(A) = First(BCD)
'd'

follow(B) = First(CD)

	first	follow( )
S → A B C D	{b, c}	{$}
A → b / ϵ	{b, ϵ}	{c}
B → c	{c}	{d}
C → d	{d}	{e}
D → e	{e}	{$}

A B C D

follow(B) = First(C D)

A B C D

follow(A) = First(B C D

C

	First()	Follow
$S \to ABCDE$	$\{a, b, c\}$	$\{\$\}$
$A \to a / \epsilon$	$\{a, \epsilon\}$	$\{b, c\}$
$B \to b / c$	$\{b, c\}$	$\{c\}$
$C \to c$	$\{c\}$	$\{d, e, \$\}$
$D \to d / \epsilon$	$\{d, \epsilon\}$	$\{e, \$\}$
$E \to e / \epsilon$	$\{e, \epsilon\}$	$\{\$\}$

$S \to ABCDE$ (e, e)

follow(C) = First(DE)

ABCDE

BCDE} follow B = First((CDE)

C

(b, c)

5 min

	first	follow
$S \to Bb / cd$	{a, b, , c}	{$}
$B \to aB / \epsilon$	{a, ϵ}	{b}
$C \to cC / \epsilon$	{c, ϵ}	{ }

C is not reachable





	First	follow
$S \to aABb$	$\{a\}$	$\{\$\}$
$A \to c/\epsilon$	$\{c, \epsilon\}$	$\{d, b\}$
$B \to d/\epsilon$	$\{d, \epsilon\}$	$\{b\}$

$aA\widetilde{B}\widetilde{b}$

$aA\cancel{B}b$

$follow(A) = First(Bb)$

d

THANK - YOU